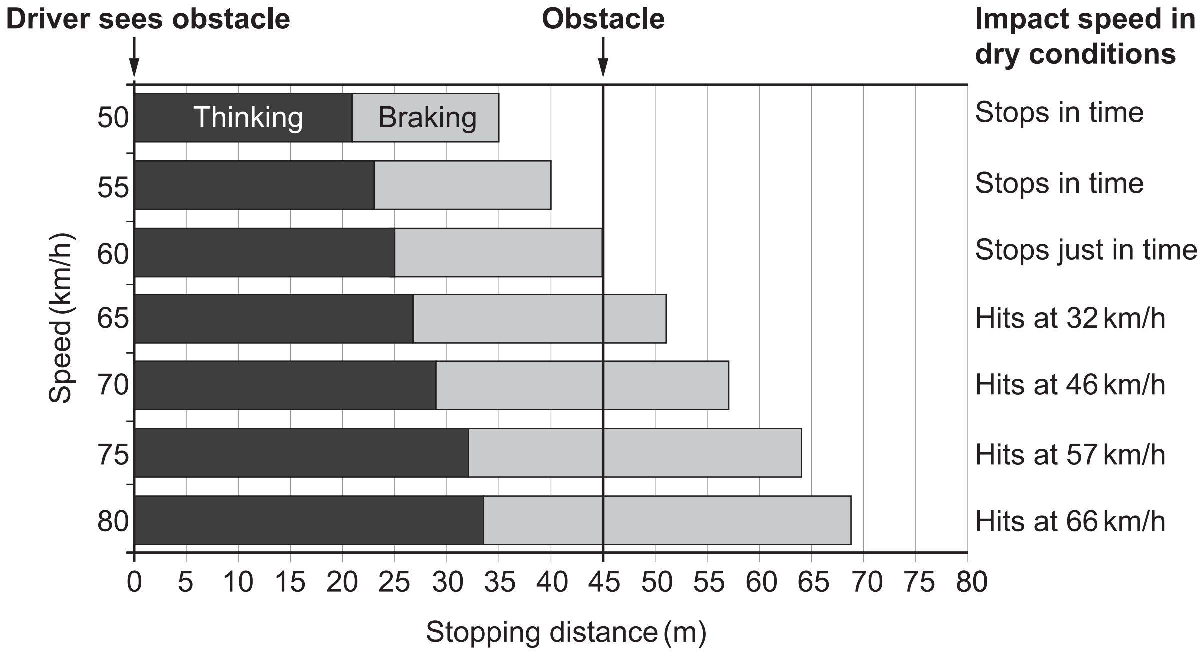
6. The chart below is used by traffic collision investigators. It gives the thinking, braking and stopping distances of cars driven at different speeds by an alert driver on a dry road.

Stopping distance is given by the following equation:

stopping distance = thinking distance + braking distance

An alert driver notices an obstacle 45 m away on the road ahead. The position of this obstacle is represented by the dark vertical line. If there is a collision, the chart also shows the impact speed with the obstacle.



(a) Use the information in the chart to answer the following questions.

(i) State the stopping distance for a speed of 50 km/h. .... m [1]

(ii) State the speed at which the car stops just in time. .... km/h [1]

(iii) State the speed which gives a braking distance of 35 m. .... km/h [1]

Examiner only

(iv) Tiredness doubles **thinking** distance for some drivers.
Gareth claims that, for these tired drivers travelling at 60 km/h, the stopping distance becomes 90 m.
With the aid of calculations, explain whether you agree with the claim. [3]

....

....

(v) Use the equation:

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

and information from the chart for a car travelling at 60 km/h (17 m/s), to calculate the **thinking time** of an alert driver. [3]

Thinking time = ... s

(b) A car is travelling at 70 km/h on a dry road when it starts to rain causing the road to become **wet**.
Complete the table below.
In each box, add either increases, decreases, or stays the same. [3]

Thinking distance	Braking distance	Stopping distance	Impact speed
....	....	increases	....

Examiner only

(c) Seat belts **and** crumple zones work together to keep the occupants of a car safe in the event of a head-on collision.
Complete the table by placing a tick (✓) in the column that matches with the action.
One has been done as an example. [2]

Action	Seat belt	Crumple zone
Increases the time of the collision		✓
Reduces force on the car		
Prevents driver continuing through the windscreen		

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